Automated Multi-Parametric MRI Segmentation of Post-Treatment Glioma Sub-Regions Using Classical Machine Learning

Midsem Report

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***Abstract*—We present two classical machine learning pipelines for binary tumor segmentation on the BraTS 2020 multi-parametric MRI dataset. Pipeline 1 combines a multi-scale Random Forest with, achieving Dice = 0.62 on a representative validation patient. Pipeline 2 employs a 53-feature-per-voxel XGBoost classifier; its best variant (v7, 1:46 class ratio, `scale_pos_weight` = 46.77) attains Mean Dice = 0.7508 and Mean IoU = 0.6269 across 136 test patients with a train-test gap of <0.002. A controlled v7-vs-v8 ablation confirms weight-based correction is superior to hard undersampling for severe class imbalance.**

***Index Terms*—Brain tumor segmentation, BraTS, Random Forest, XGBoost, multi-parametric MRI, Dice, class imbalance.**

I. Introduction

Problem.

Gliomas are among the most aggressive forms of primary brain tumors. Numerous times, patients will have surgeries performed with chemotherapy/radiation treatment, but the tumor continues to grow back [1]. Due to the fact that working with gliomas has a high recurrence rate, follow-up MRIs are used to evaluate these patients after surgical treatment or chemotherapy and radiation. Due to the complexity of interpreting these MRIs (and the fact that the surrounding surgery cavity, radiation impact, or necrotic area can easily resemble active tumors), it can be difficult for physicians or radiologists to determine if there is, in fact, an active tumor. For this reason there is a need for automated and repeatable ways to identify the location of tumor within the patient's MRI images.

The challenge is voxel-level classification, in other words classifying the thousands of small (3-D) units of the patient's brain image in 3D space. Each of the 3D voxel units must have a classification of either being a tumor (1) or non-tumor tissue(0). The determination of which of the voxel units are tumor and non-tumor is based on the four spatially aligned MRI modalities: $\{T1, T1ce, T2, \text{FLAIR}\}$each of which highlight different aspects of the tumor structure.

Classifying tumor voxels is exceedingly difficult because less than 2% of the total brain voxels,of all the brain voxels are tumor voxels. In addition to the major imbalance of voxel categories, MRI intensity values can widely differ among the patient population and there are approximately 1.7 million voxels per scan making the problem of classifying gliomas an extremely computationally expensive problem.

Dataset. BraTS 2020 [1]

1) The dataset provides skull-stripped MRI volumes of size $240 \times 240 \times 155$ stored in NIfTI format.
2) Expert annotations are provided for three tumor sub-regions:
   - NCR/NET (Necrotic and Non-Enhancing Tumor Core) – label 1
   - Edema – label 2
   - Enhancing Tumor – label 4
3) For this study, all tumor labels are merged to form a binary segmentation task.
4) The dataset is split into training, validation, and test sets as follows:
   - 75% Training
   - 15% Validation
   - 10% Testing
5) The dataset contains MRI scans from a total of 136 patients.

Preprocessing.

1) Each patient volume is first cropped to a tight bounding box containing only the non-zero brain voxels. This reduces the overall volume by approximately $\sim 60\%$.
2) Z-score normalisation is then applied separately to each MRI modality using only the brain voxels, computed as:

$$V_{\text{norm}}(v) = \frac{V(v) - \mu_\Omega}{\sigma_\Omega}$$

3) Finally, the processed volumes are saved as `.npy` files.
4) A binary brain mask M is used to ensure that all feature extraction is performed only on the non-zero brain voxels.

II. Methodology

A. Pipeline 1 — Random Forest

Feature extraction. Multimodal MRI volumes (T1, T1c, T2, FLAIR) are used to extract a total of 51 features from each superpixel. Voxel features are made up of 4 groups of features based on intensity, gradient, Laplacian, Laplacian of Gaussian; these features are calculated at 3 scales for each modality, leading to a total of 24 voxel features. From the voxel features of each superpixel, we will compute both the mean and standard deviation for each voxel feature set, providing us with 48 voxel feature statistics. Additionally, we will calculate 3 sets of local texture statistics for each superpixel, yielding a final total of 51 features for each superpixel.

Balanced sampling. In order to create a balanced dataset, we will resample the superpixel samples from both the tumor and background classes and pool them together to create one training set (X_{train}, y_{train}).

Model training. A RandomForest classifier (n=120, depth=20) is trained on the balanced dataset.

Segmentation. On inference, superpixel features are computed from the segmented images, the classifier is subsequently used to assign labels to the superpixels, and the final segmentation result will be obtained by back-projecting superpixel labels to voxel space.

Evaluation. Segmentation accuracy is measured using the Dice similarity coefficient.

Algorithm 1 Brain Tumor Segmentation using Superpixel Features and Random Forest

```
Load dataset of multi–modal MRI volumes
Split patients into Train, Validation, Test
for each patient in Train set do
    Load T1, T1c, T2, FLAIR and GT
    Normalize intensities
    Downsample volume
    Extract voxel features
    Apply 3D SLIC superpixel segmentation
    Compute superpixel features (mean, std, texture)
    Assign label using majority voting
    Store features and labels
end for
Balance dataset using resampling
Train Random Forest classifier
for each patient in Validation set do
    Load MRI volumes
    Extract features
    Apply SLIC
    Predict labels using RF
    Convert to voxel segmentation
    Compute Dice score
end for
for each patient in Test set do
    Load MRI volumes
    Extract features
    Apply SLIC
    Predict labels using RF
    Convert to voxel segmentation
    Compute Dice score
end for
Output mean Dice score
=0
```

B. Pipeline 2 — XGBoost (53 Features per Voxel)

Feature extraction. Per-modality features (11×4=44): raw intensity (1); local mean at $3^3, 5^3, 9^3$ (3); local std at $3^3, 5^3, 9^3$ (3); Sobel gradient magnitude (1); Gaussian blur σ=1, 2 (2); Difference-of-Gaussians (1). Global features (9): normalised spatial coords x, y, z (3); cross-modal ratios/diffs for all $\binom{4}{2}$=6 pairs (6).

1) **Sampling**
   - **v7:** Uses the natural class imbalance ratio of $1:46$ with `scale_pos_weight` $= 46.77$.
   - **v8:** Applies hard undersampling of background voxels to obtain a $1:10$ ratio.
2) **Training**
   - The XGBoost model is trained on approximately 2.07 million voxels.
   - Log-loss is monitored at each boosting round using a held-out validation set.
3) **Inference**
   - A voxel is classified as tumor if

$$\hat{y}(v) = \mathbf{1}[P(\text{tumor} \mid v) \geq 0.6].$$

Algorithm 2 Pipeline 2 — XGBoost 53-Feature

Require: $\{T1, T1ce, T2, \text{FLAIR}\}$ volumes, seg. mask
Ensure: Binary tumor mask

```
for each patient in train set do
    Crop to brain bbox; Z-score normalise; build mask M
    for each voxel v ∈ M do
        Per-modality (11×4=44): intensity, mean×3, std×3, grad, blur×2, DoG
        Global (9): coords x, y, z; cross-modal ratios/diffs ((4 2)=6)
    end for
    if v7 then keep 1:46 ratio; scale_pos_weight= 46.77
    else if v8 then undersample background to 1:10
    end if
    Append to X_train, y_train
end for
Train XGBoost(n=300, depth=8, lr=0.1, sub=0.8, binary:logistic, spw= 46.77)
Monitor logloss on val set each round
Save model xgb_v7.pkl; threshold best_thr.npy= 0.6
for each test patient do
    Extract 53 features for all brain voxels (batched)
    ŷ(v) ← 1 if P(tumor|v) ≥ 0.6
    return binary tumor mask
end for
Compute Dice, IoU, AUC-ROC per patient; save to CSV =0
```

III. Results

A. Evaluation Metrics

$$\text{DSC} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \qquad \text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}$$

AUC-ROC is also used as a performance evaluation metric.

B. Pipeline 1 — Random Forest

(patient BRATS-GLI-02269-100)

- Avg Dice Score: 0.61
- Validation Dice: 0.62
- Dice Score(for the image given): 0.8529
- Reduced isolated false positives in segmentation output

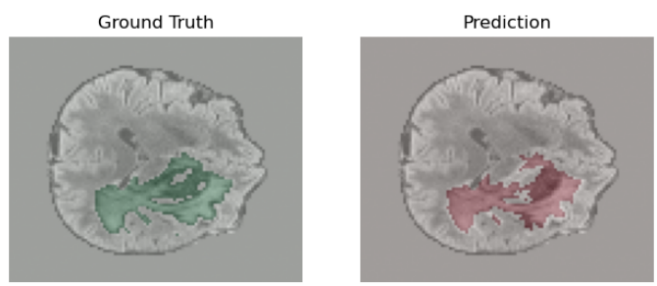

Fig. 1: Pipeline 1 segmentation results on patient BRATS-GLI-02069-101. (Top row) Input modalities: T1, T1ce, T2, FLAIR. (Middle row) Ground truth segmentation and predicted probability map. (Bottom row) Final segmentation after Superpixel aggregation and testing shows spatially coherent tumor boundaries.

C. Pipeline 2 — XGBoost v7

(136 test patients)

- Mean Dice: 0.7508
- Mean IoU: 0.6269

- AUC-ROC: 0.9897
- Best patient: BraTS-GLI-02141-100 (Dice = 0.9452)
- Worst patient: BraTS-GLI-02767-100 (Dice = 0.0000 — very small/atypical lesion)
- Train Dice: 0.7494
- Test Dice: 0.7508
- Train–Test gap: -0.0014 (indicates good generalization)

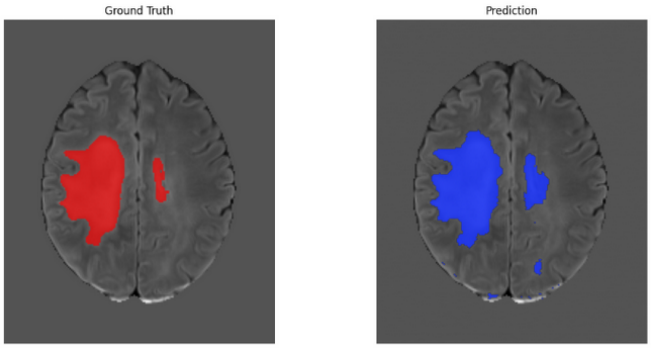

Fig. 2: Pipeline 2 (XGBoost v7) typical-case segmentation.

D. Pipeline 2 — XGBoost v8

(136 test patients)

- Mean Dice: 0.6282
- Mean IoU: 0.4874
- Train Dice: 0.6216
- Test Dice: 0.6282
- Train–Test gap: -0.0066

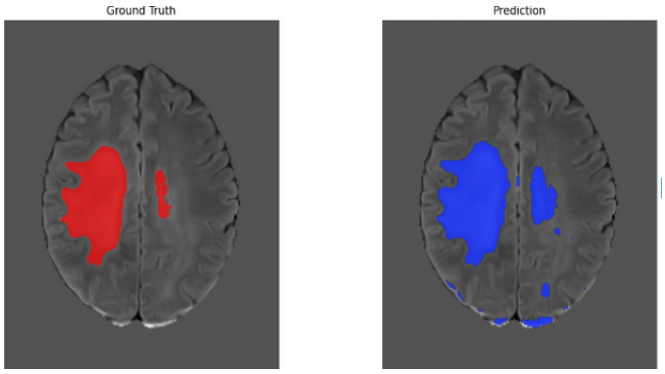

Fig. 3: Pipeline 2 (XGBoost v8) typical-case segmentation.

v7 vs. v8 comparison is shown in Table I.

TABLE I: Pipeline 2 Test Results — v7 vs. v8 (136 Patients)

Metric	**v7 (1:46)**	**v8 (1:10)**
Mean Dice	**0.7508**	0.6282
Std Dice	0.1764	0.2041
Median Dice	**0.8046**	0.6701
Mean IoU	**0.6269**	0.4874
Dice ≥ 0.9	**104/136**	2/136
Dice ≥ 0.7	**104/136**	61/136
Dice ≥ 0.5	**125/136**	104/136
Train Dice	0.7494	0.6216
Train-Test Gap	-0.0014	-0.0066
Model Size	1.7 MB	**0.3 MB**

IV. Discussion

Discussion:

Pipeline 2 surpasses Pipeline 1 by $+0.167$ Dice points.The gain arises from: (i) it uses more richer 53-feature vector with three neighbourhood scale terms and cross-model interaction term not used in Pipeline 1; (ii) it uses iterative residual correction of gradient boosting; (iii) `scale_pos_weight` of model-level imbalance correction without the cleanup of background context.

This is testified by the v7-vs-v8 ablation where hard undersampling in v8 removes most of the background examples and the removal of critical ones healthy-tissue setting and decreasing Mean Dice by -0.123. It is also worth mentioning that there was a bug in the sampling in the early v7 code which moved the effective ratio from 1:46 $\approx$ 1:10.3 (air/skull voxels are treated as background), increased the effective ratio of 2.1% to 9.75% of effective tumor fraction and increased Dice of estimated between 0.65–0.68 to 0.75+, the sensitivity of boosted classifiers to choice of class ratio.

Failure case analysis. Of 11 patients with Dice <0.5: (i) seven have very less lesion volumes ($<5\,000$ GT voxels) making DSC ill-conditioned; (ii) two of them have T2-hypointense necrotic cores that are not well modeled in training; (iii) two of them have post-operative resection cavities that have similar FLAIR signal to that of edema and is therefore generates large false-positive regions.

Future scope. Future work: Multi-label sub-region segmentation (NCR/ NET, Edema, ET) one-vs-rest XGBoost: The use of GLCM texture features to better distinguish boundaries [5]; Chan-Vese active contour: post-processing [6]; lesion-size-aware adaptive thresholding; a hybrid cascade with the speed of Pipeline 2 and the spatial resolution of Pipeline 1.

V. Conclusion

An evaluation of two classical segmentation methods that use only a CPU was performed with 136 BraTS 2020 patients. Pipeline 2 (XGBoost, 53 features per voxel) achieved Mean Dice = 0.7508 at 1.7 MB model size, substantially exceeding Pipeline 1 (Random Forest, Dice = 0.61). Both pseudocodes were presented in full. A comparative evaluation of v7-vs-v8 was done to show that using `scale_pos_weight` versus hard undersampling provides a major improvement for highly unbalanced classes. Failure-mode analysis identified three distinct causes of low Dice. Week 6 will target multi-label subregion extension and failure-mode mitigations.

References

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